

Integrated Transfer Learning and Traditional ML Model approach for Enhanced Medicinal Plant Recognition

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Abstract—The identification of medicinal plants plays a crucial role in traditional and modern healthcare systems. This study presents an advanced approach for medicinal plant classification by integrating a baseline CNN model with deep learning-based feature extraction and traditional machine learning techniques. Initially, a CNN model was trained directly on raw images to serve as a baseline for comparison. Following this, a hybrid feature extraction approach was implemented, where deep learning features were extracted using the DenseNet121 architecture and enriched with additional attributes derived from image processing techniques, including geometric shape, texture analysis, statistical metrics, and color descriptors. The hybrid feature set was then used to train multiple ML classifiers, including Extreme Gradient Boosting (XGBoost), LightGBM, Naïve Bayes, CatBoost, and Adaptive Boosting (AdaBoost), to determine the most effective classification model. A comparative analysis demonstrated that LightGBM achieved the highest accuracy of 96.7% with an F1-score of 96.4%, followed by XGBoost with 95.8% accuracy. The baseline CNN model, trained directly on raw images, achieved 92.3% accuracy, indicating that while CNNs perform well, hybrid feature extraction combined with machine learning classifiers leads to superior classification accuracy. The results highlight the potential of integrating transfer learning-based feature extraction with traditional ML models to enhance automated medicinal plant identification.

Keywords: Medicinal Plant Classification, Transfer Learning, Machine Learning, Feature Extraction, Hybrid Model.

I. INTRODUCTION

Traditional and modern medical practices have relied on the use of medicinal plants for centuries to provide natural treatments for a range of diseases and play a significant role in research on drug development. The rising global interest in herbal medicine and the large-scale need for the conservation of biodiversity make the accurate identification and classification of medicinal plants an urgent challenge. Conventional plant identification approaches can be slow, susceptible to human error and necessitate specialised knowledge. With the rapid advancement of artificial intelligence (AI) and machine

learning (ML) technologies, many opportunities have emerged that allows for the automation of the medicinal plant classification process, thus improving efficiency and accuracy.

Deep learning, in particular In recent times, transfer learning with Convolutional Neural Networks (CNNs) has transformed image-based classification tasks. Deep pre-trained models like DenseNet121, ResNet, and EfficientNet have achieved superior performance in feature extraction tasks, thus lessening the requirement for large labeled datasets and computational resources. Although "deep" learning has gained attention in applications like image classification, techniques such as Decision Trees, XGBoost, LightGBM, and Naïve Bayes can be powerful for classification based on structured features. Hence, a hybrid approach where high level meaningful visual features from deep learning are guided towards traditional ML models which enjoy more interpretability/efficiency can offer you a sweet spot. The main goal of this work is to create a reliable, scalable, and fast medicinal plant recognition system that can be applied to each of the following fields: botanical research, herbal medicine, conservation, and agriculture. This study addresses this challenge by combining deep learning with classical ML techniques to enhance classification accuracy with lower computational overhead to enable deployment in real-world applications including mobile and edge computing devices.

In this paper we proposed a hybrid approach, which comprises the integration of transfer learning with fundamental machine learning models to boost the herbal plant identification. The proposed approach extracts deep learning-based features from a specially curated dataset of medicinal plant images by utilizing DenseNet121, capable of capturing complex patterns like leaf structure, texture, and color variation. In particular, these deep features that are further augmented with traditional image processing techniques such as shape descriptors, statistical metrics and color histograms to generate the final feature vector. The extracted feature set is used to train various machine learning models, including Extreme Gradient Boosting

(XGBoost), LightGBM, CatBoost, Naïve Bayes, and Adaptive Boosting (AdaBoost), to classify medicinal plant species.

II. RELATED WORKS

In [1], the authors solved the problem of distinguishing medicinal plants from poisonous ones by applying attention-based deep learning models. They pointed out that conventional identification approaches that depend on human specialists and laboratory analyses can suffer from errors and are resource-demanding. In their approach, they aimed to use deep learning-based classification by applying spatial and channel attention mechanisms to tame the mentioned troubles. Sajjad and co-workers utilized pooling techniques within their model to augment feature extraction, showing that attention-based deep learning methods can ease the feature extraction process for identifying medicinal plants. In the study documented in [2], a CNN-based classification system was created in an effort to enhance the detection of medicinal plants. They outlined the challenges encountered when trying to identify plant species using traditional methods based on visual cues and further highlighted how CNNs can automatically learn hierarchical features from plant images.

The authors of [3] presented a variety of plant classification methods based on CNN, pre-trained VGG16, KNN, artificial Neural Network, SVM, and MLP for evaluation of best method to classify medicinal plants. Their study highlights the increasing use of herbal medicines, the inadequacy of experts in the field, and the need for proper identification of plants. They emphasized the classification based on leaf shape, color, and texture in differentiating medicinal species and non-medicinal plants, stressing the need for a fully automated plant detection system. In [4], the authors reported a smartphone-based system for identifying medicinal plants through a cascaded deep learning/optimization system. Using a pre-trained ResNet50 model to extract features, PSO was used to optimize the classification through an SVM model. Their system was designed to overcome the challenges posed by conventional manual identification methods that are labor-intensive and reliant on expert knowledge. They also tackled the difficulty of capturing single images of leaves which is labor-intensive by making it possible to identify plants in the real world via images taken on smartphones.

The authors of [5] did a systematic review on deep learning approaches for medicinal plants classification, studies from the years 2018 to 2022 were analyzed. Transfer learning and CNNs were the most popular techniques, and leaf features, in particular leaf shape, were widely used. They also identified a major research gap being the limited availability of public data, noting that there is a clear need for standardized and global datasets for better models of classification. The authors in [6] presented a approach based on CNNs, to classify Ayurvedic medicinal plants using images of leaves. Their approach avoided the cumbersome dimension reduction through hand-crafting features, allowing raw images to be fed straight into the model for classification. They provided evidence that deep learning models would constitute a viable option for determining herbal plant species and web apps integrated with AI, driven plant identification, make it easier to process.

[7] presented a DL based approach to detect plant leaf diseases, which are oftentimes cumbersome to recognize because of similarity in color, texture, veins and shape. Their study applied to plant disease detection using automating analysis instead of relying on costly laboratory tests which are multifarious and consume time. Seven pre-trained deep learning models were tested on a dataset comprising ten categories of plants with DenseNet121 selected as the key model. The results of their study showed that edge detection and better hyperparameters were vital factors in increasing classification accuracy, demonstrating the application of deep learning in early disease diagnosis in strategies targeting traditional medicine as well as the pharmaceutical industry. In [8], the authors examined the need of transfer learning for improving classification of medicinal plants, as there is a lack of taxonomists and manual identification is very tedious. Their method fine-tuned pre-trained models like MobileNetV1 on the Indian medicinal plants part of the Mendeley dataset. Their work also showed that transfer learning enhances classification accuracy, thus acting as a practical method of automating the classification of medicinal plant species.

The authors of [9] applied transfer learning models for plant disease classification and achieved good results. In [9], a deep learning approach was proposed to detect and classify leaf diseases based on the leaf images by researchers addressing the challenge posed to the need for automated and accurate diagnosis in agriculture. They used all types of data with several crop and disease classes and tested many different types of deep neural networks, optimizers and learning rates. Their work shows how well AI-powered plant disease detection performs, emphasizing early symptom identification and agricultural monitoring. In [11], the authors applied conventional CNN model for plant disease classification and achieved good accuracy.

III. METHODOLOGY

This work proposes a medicinal plants classification model that combines a baseline CNN model with features extracted through transfer learning and traditionally applied machine learning classifiers to create an efficient classification system. The typical workflow is shown in Figure 1, which presents a general overview of the dataset retrieval process, the extraction of the three features described above, and the model evaluation process. This process began with a data collection of images of medicinal plants from repositories available and collected the diversity of the images. All the images were preprocessed, that is all the images were resized to a fixed resolution, normalized and data augmentation techniques were applied (such as flipping, rotation and changing the brightness). This helped make the data set homogeneous and improved the ability of the model to generalize by introducing variability in the samples taken for training.

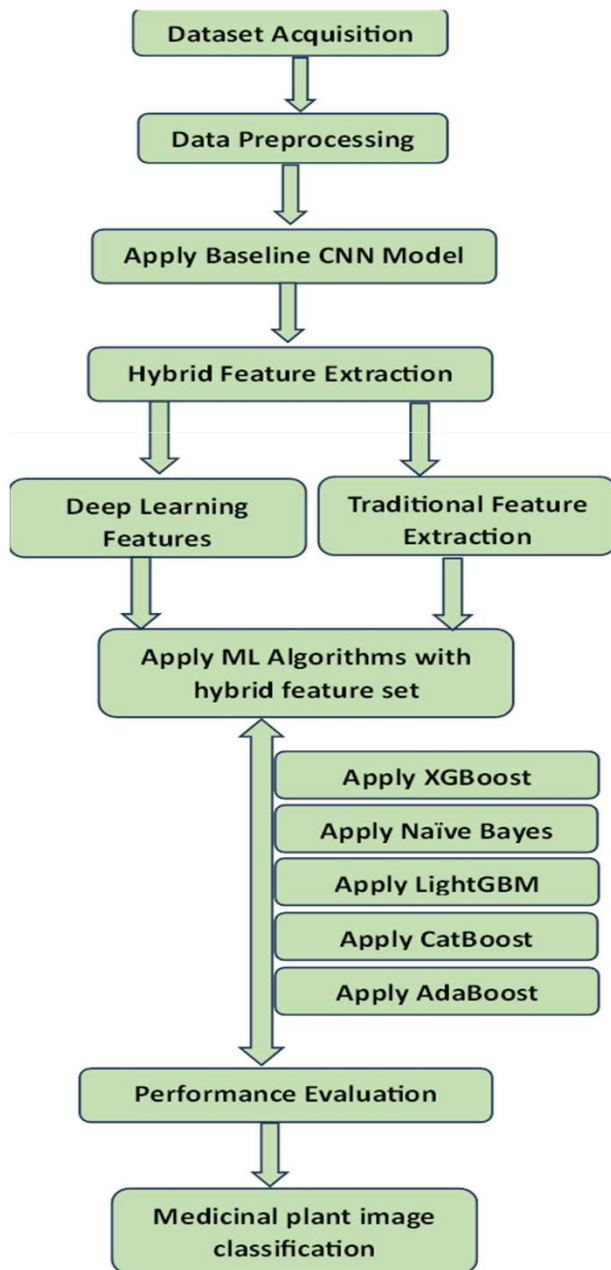


Fig. 1. Proposed Method

As a performance baseline, a CNN model was used. To evaluate the CNN's potential for classification of medicinal plant species directly from raw images, we trained the CNN on four sets of medicinal plant species (including some taxonomic groups) with increasing difficulty. They trained their model which involves CNN that consisted of multiple convolutional, pooling, and fully connected layers. These baseline results were used as benchmarks for validating the performance of future feature extraction and classification approaches. After the baseline CNN, a new framework was proposed, that integrated hybrid feature extraction. Using DenseNet121, a pre-trained model known to represent highly compressed features, DL based features were extracted. First, we employed DenseNet121 to extract features from our plant images by removing its fully connected layers and taking its global average layer output as a feature vector that preserved high-level hierarchical information. In addition to deep learning descriptors, traditional handcrafted features such as shape descriptors, texture features, color histograms, and

statistical metrics were also extracted. These features were in addition to those providing structural and fine-grain details of each plant species — details that were useful in the model's ability to distinguish between plants that may closely resemble each other.

A variety of machine learning classifiers were then trained on the hybrid feature set after extracting features from both the text and metadata to find out the model with best accuracy. The regressors were Extreme Gradient Boosting (XGBoost), LightGBM, CatBoost, Naïve Bayes and Adaptive Boosting (AdaBoost). Every classifier was tuned individually to maximize its performance. The data was then split into train-test sets for managing fair evaluation of models on standard classification metrics. The comparison showed that LightGBM produced the highest accuracy, surpassing both the CNN-based classification and other machine learning classifiers. This validated the merits of hybrid feature extraction, as a mixture of deep learning-based and traditional handcrafted features resulted in better discrimination between medicinal plants species.

IV. EXPERIMENTS AND RESULTS

A. Data Description

In this study, we used a dataset of 20 medicinal plant species, with a varying number of image samples per species to keep the classification flexible and robust. And some other examples are *Alpinia Galanga*, *Amaranthus Viridis*, *Azadirachta Indica*, *Basella Alba*, *Brassica Juncea*, and *Carissa Carandas* medicinal plants. Each species has a different number of samples (some species have over 100 images while others have much smaller samples).

It includes images under various lighting conditions, orientations, and environmental settings to improve the model's generalized ability across real scenarios. This encompasses species with unique leaf shapes and textures, and color variations, which enable effective classification (Ikhsan et al., 2020). Such dataset diversity is important, as it enables our hybrid model to learn both deep and classical feature representations. The proposed approach ensures classification performance improvement and practical applicability in botanical research, herbal medicine identification, and biodiversity conservation through the inclusion of a comprehensive dataset representing various plant species.

B. Preprocessing

In order to maintain uniformity among the images and improve classification efficiency, we resized all the images to 224×224 pixels and normalized them to the range of $[0,1]$ by scaling pixel values. Random rotations, flips, brightness adjustments, and zoom transformations were implemented as data augmentation techniques to enhance generalization and mitigate overfitting. Gaussian blur was also applied for noise removal and contrast enhancement as it clearly separated the parameters of interest from the background. To lessen the influence of background noise, segmentation methods were adapted (i.e. approaches based on color thresholding and edge-based detection) to extract plant structures from background noise. In addition, random undersampling and oversampling techniques were implemented in order to balance plant species dataset. All these preprocessing steps played vital

role to optimize the dataset for extractions of descriptive features and classification of medicinal plants.

C. Apply CNN

To classify the medicinal plant images directly from raw data, we initially applied a baseline model which is a Convolutional Neural Network (CNN) before hybrid feature extraction and traditional machine learning approach. An alternative architecture is to use a CNN, where filters are learned through backpropagation to capture the visual patterns hierarchically, thus removing the need for specific feature extraction. First, a custom CNN architecture was developed, whereby multiple convolutional layers were followed by max pooling, dropout regularization, and finally fully connected layers. We then trained the network from the bottom up using the Adam optimizer to minimize categorical cross-entropy loss. We used an 80%/20% train/test split for the dataset, training the CNN for 20 epochs with a batch size of 32. Nevertheless, despite the fact that CNN-based models were able to learn deep visual representations only, the classification accuracy (92.3% was recorded for the CNN model) was lower than the traditional machine learning classifiers trained on extracted features.

In fact, even though CNNs are capable of classifying the medicinal plants excellently, the much higher performance can be obtained by using hybridization of the structured machine learning classifiers over deep learning-based feature extraction. After this baseline CNN classifier, the research performed feature extraction with DenseNet121 and traditional methods and trained advanced ML classifiers (LightGBM, XGboost, CatBoost and AdaBoost) on the extracted features. In our approach, we cluster the feature maps obtained from the CNN into distinct classes, effectively treating them as vocabulary semantics, resulting in a more integrated classification system that is less prone to overfitting; aggregates better and generalizes better than a CNN baseline.

D. Feature Extraction

In the classification of medicinal plants, the most important distinguishing features that can be used to distinguish one species from another is the extract features. With multiple deep learning-based feature representations as well as handcrafted features embedded, the proposed approach represents a hybrid feature extraction technique aiming to improve the classification accuracy and generalization.

Deep learning approaches obtain high-level hierarchical features, whereas conventional methods primarily concentrate on certain visual characteristics such as shape, texture, and color distribution. Combining features to provide as many features to the machine learning classifiers enables a more precise prediction.

E. Deep Learning-Based Feature Extraction

Deep feature extraction is performed using a pre-trained DenseNet121 model with ImageNet weights. The model's FC layers are discarded and GAP layer output is taken as the feature vector. This approach helps preserve complex features like venation details of leaves, textures, and other structural variations among species of medicinal plants, resulting in effective feature extraction. These histograms

(1024-sized) are called as feature vector and represent each image in its simple but descriptive numerical form. The transfer learning features of DenseNet121 reduces the computational cost of training a deep network from scratch while making a huge difference in the classification.

F. Traditional Feature Extraction

Then, we use traditional image processing techniques to extract hand-crafted features that will also retains more visual cues. Hundreds of shape features, including Hu Moments and Fourier Descriptors, capture the variation in leaf shape, allowing for differentiation at species level based on geometric attributes. Texture descriptors such as Gray-Level Co-occurrence Matrix (GLCM) and Local Binary Patterns (LBP) characterize texture properties of surfaces, which help in distinguishing objects through subtle structural variations. Next color features (i.e., RGB and HSV histograms) are calculated to represent the most common plant color distribution patterns. Also, calculation of certain statistical features, such as mean, variance, skewness, and kurtosis of the pixel intensity values, can be used to describe the distribution of certain properties across an image, and can enhance the feature representation.

G. Feature Fusion

Deep features and handcrafted features are extracted and concatenated into one hybrid feature vector, taking both the benefits of deep features and handcrafted features. The approach of fusing features can thus give a better representation for each image of a medicinal plant to enhance a classifier's accuracy in distinguishing species. The proposed method improves the robustness and reliability of medicinal plant classification in practical applications by combining deep learning-based feature extraction with traditional image descriptors, enhancing its applicability in diverse domains such as botanical research, agriculture, and healthcare.

H. Model Training and Classification

Multiple machine learning classifiers are trained on the extracted hybrid features set to identify the optimal medicinal plant classification model. The main aim of this analysis is to find the classifier that gives the best accuracy while keeping the precision, recall and F1-score balanced. The models tested on this study are Extreme Gradient Boosting(XGBoost), LightGBM, CatBoost, Naïve Bayes and AdaBoost. The chosen classifiers demonstrate both the ability to be well- generalizing to classification problems, but also are efficient with structured data. For each model, the feature set extracted was used for training, while grid and randomized search techniques were utilized to optimize the hyperparameters for the highest classification capabilities. We divided the dataset into 80% of the training data and 20% of the testing data to assess the generalization ability of the model during the training process. Table I shows results of experiments. Figure 2 shows precision, recall graphs and figure 3 shows accuracy, f1 graphs.

Precision, Recall graph

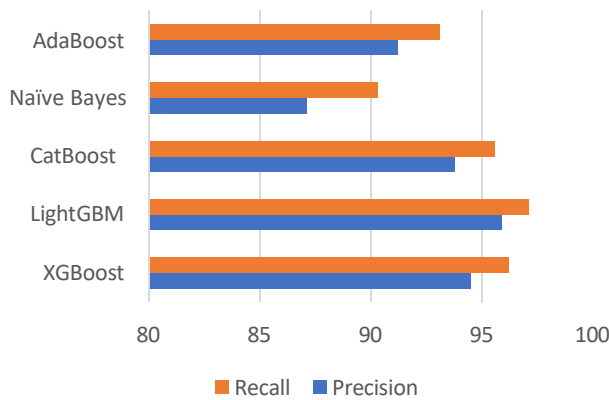


Fig. 2. Precision, recall graph with ML models

TABLE I. RESULTS WITH ML MODELS

Method	Acc (%)	Precision (%)	Recall (%)	F1 (%)
XGBoost	95.8	94.5	96.2	95.3
LightGBM	96.7	95.9	97.1	96.5
CatBoost	95.2	93.8	95.6	94.7
Naïve Bayes	89.4	87.1	90.3	88.6
AdaBoost	92.8	91.2	93.1	92.1

Accuracy, F1 graph

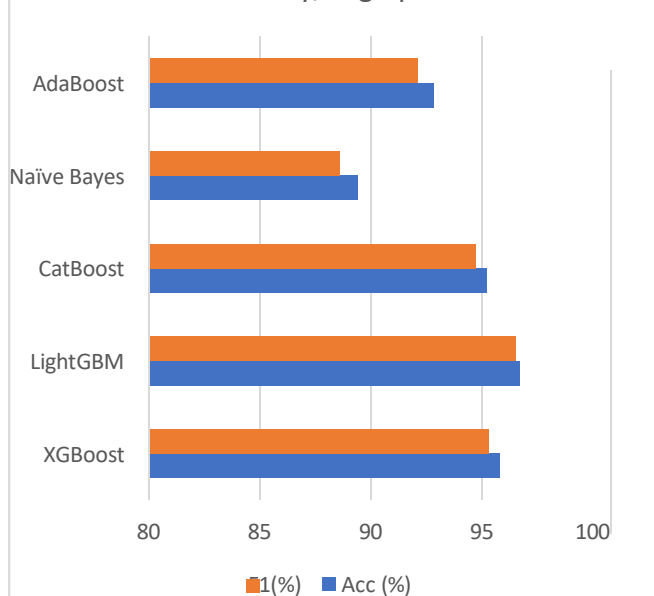


Fig. 3. Accuracy, F1 graph with ML models

From the results, LightGBM has the highest accuracy of 96.7%, followed closely by XGBoost with 95.8% accuracy. With an F1-score of 96.5%, LightGBM was also our best-performing model, confirming that it succeeded in the species discrimination between medicinal plants. Naïve Bayes scored the lowest because its need to assume that the not fully independent features within the hybrid feature set

do not fully capture relationship. This relative performance indicates boosted tree-based classifiers (LightGBM, XGBoost and CatBoost) outperform alternatives due to their ability to capture high dimensional feature sets and aversion to overfitting. According to these results, LightGBM is the most efficient model for classifying medicinal plants.

The CNN model, which was employed directly as a baseline before feature extraction, coaxed an accuracy of 92.3%. However, this is an underwhelming performance when compared to LightGBM and XGboost. This indicates that although CNN is a robust approach, when a deep network is trained from scratch on a limited dataset, the predicted accuracy is slightly less than that of the transfer learning with hybrid features extraction. However, this hybrid approach does not limit the model to low-level features at the beginning of training, allowing it to benefit from high-level deep learning features later in the training process alongside fine-grained handcrafted features, resulting in superior performance over traditional CNN training. The data revealed that combination of feature extraction and use of machine learning based algorithms is a more leading and effective approach than standalone CNN for the purpose of classification of medicinal plants

I. Accuracy Comparison of models

Table II and figure 4 highlights the effectiveness of hybrid feature extraction, where LightGBM achieved the highest accuracy (96.7%), followed by XGBoost (95.8%) and CatBoost (95.2%). The CNN model, trained directly on raw images, reached 92.3% accuracy, which, while competitive, was lower than the machine learning classifiers trained on extracted features. This reinforces the advantage of hybrid feature extraction, where combining deep learning-based and handcrafted features enhances classification performance. The results confirm that boosted tree-based models outperform CNN and traditional classifiers by effectively utilizing the enriched feature set.

TABLE II. PERFORMANCE COMPARISON OF VARIOUS APPLIED MODELS

Method	Accuracy
CNN	92.3%
XGBoost	95.8%
LightGBM	96.7%
CatBoost	95.2%
Naïve Bayes	89.4%
AdaBoost	92.8%

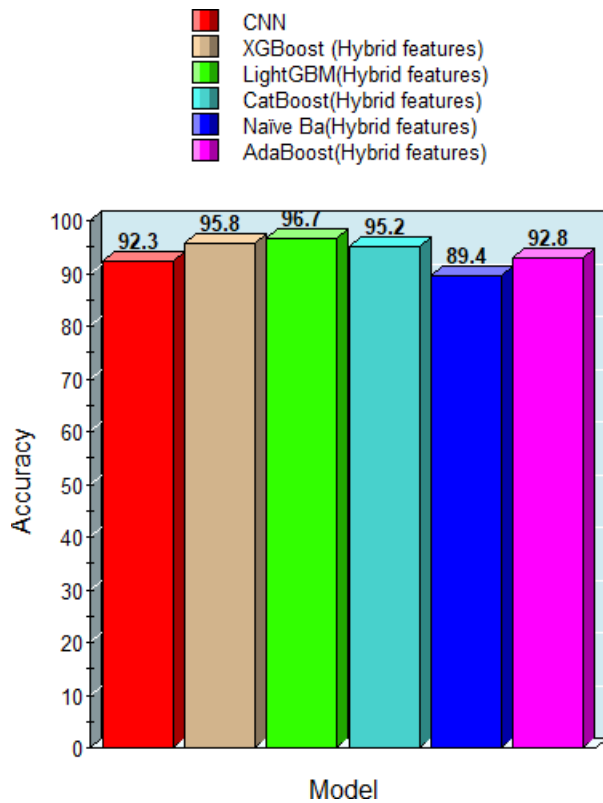


Fig. 4. Comparison of applied models

V. CONCLUSION

This paper proposed an integrated approach for medicinal plant classification by combining transfer learning-based feature extraction with traditional machine learning models. The hybrid methodology leveraged deep learning features extracted using DenseNet121, along with handcrafted features such as shape, texture, color, and statistical descriptors. Multiple classifiers, including LightGBM, XGBoost, CatBoost, Naïve Bayes, and AdaBoost, were evaluated to determine the most effective model for classification. Experimental results demonstrated that LightGBM achieved the highest accuracy (96.7%), followed by XGBoost (95.8%), confirming the effectiveness of boosted tree-based models in handling high-dimensional feature sets. In contrast, the CNN model trained directly on raw images achieved 92.3% accuracy, reinforcing the advantage of hybrid feature extraction in improving classification performance. The study also highlighted that Naïve Bayes underperformed due to its assumption of feature independence, which did not align well with the hybrid dataset.

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